

Yield Curve Equations for M-Library

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1 Abstract

This paper explains various equations to build yield curve for each currency market in the M-Library. Firstly, it introduces product equations which are basis of each yield curve, and secondly, it explains par rate equations which are used in curve calibration processes in the M-Library. (The explanation for how to use short market (FRA, Futures, etc.) in yield curve is beyond the scope of this paper.)

Please note that we have made the following 3 assumptions for building curves.

1) Forward Rate is constant against collateral. 2) Forward FX is constant against collateral. 3) Xccy Basis Swap is USD collateral.

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2 Notation

- i : index of payment dates scheduled by Quarterly (0,1,2,...,4n)
- $4n$: swap maturity (1Y,2Y,...)
- τ^{XXX} : day count fraction by XXX(30/360,ACT/365,...)
- S^{XXX} : fixed Swap rate against XXX(3M Libor,6M Libor,...)
- B^{XXX} : XXX(3M-6M,Libor-OIS,...) Basis swap spread
- L^{XXX} : XXX(3M,6M,...) forward Libor rate
- O^{XXX} : XXX(Compounding/Averaging) forward OIS rate
- DF : Discount Factor of (target currency's) OIS curve
- DF^{XXX} : DF of (target currency's) XXX(3M Libor,Xccy,...) curve¹
- $DF^{[AAA]}$: DF of [AAA] OIS curve ([AAA]: any currency)
- $DF^{[AAA]on[BBB]}$: DF of [AAA] discount curve on [BBB] OIS curve ([AAA], [BBB]: any currency)
- FX_0 : Spot FX (against USD)
- FX_i : forward FX (against USD) at i^{th} payment date
- Cov : deterministic log-normal Covariance between forward FX (against USD) and USD forward bond²

¹We assume Xccy (Basis) Swap is USD collateral, therefore DF^{Xccy} is DF of discount curve on USD OIS curve.

²For further details, please refer to [1].

3 USD Market

3.1 Product Equation

1. Overnight Index Swap($\leq 2Y$):

$$S^{OIS} \sum_{i=1}^n \tau_{4(i-1)}^{ACT/360} DF_{4i} = \sum_{i=1}^n O_{4(i-1)}^{Cmpd} \tau_{4(i-1)}^{ACT/360} DF_{4i} \quad (1)$$

2. Libor-OIS Basis Swap($> 2Y$):

$$\sum_{i=1}^{4n} (O_{i-1}^{Avg} + B^{LO}) \tau_{i-1}^{ACT/360} DF_i = \sum_{i=1}^{4n} L_{i-1}^{3M} \tau_{i-1}^{ACT/360} DF_i \quad (2)$$

3. 3M Libor Swap:

$$S^{3ML} \sum_{i=1}^{2n} \tau_{2(i-1)}^{30/360} DF_{2i} = \sum_{i=1}^{4n} L_{i-1}^{3M} \tau_{i-1}^{ACT/360} DF_i \quad (3)$$

4. 3M-6M Basis Swap:

$$\sum_{i=1}^{2n} [\prod_{j=1}^2 \{1 + (L_{2i-3+j}^{3M} + B^{36}) \tau_{2i-3+j}^{ACT/360}\} - 1] DF_{2i} = \sum_{i=1}^{2n} L_{2(i-1)}^{6M} \tau_{2(i-1)}^{ACT/360} DF_{2i} \quad (4)$$

3.2 Par Rate Equation

Note³

1. Overnight Index Swap($\leq 2Y$):

$$S^{OIS} = \frac{\sum_{i=1}^n \boxed{O_{4(i-1)}^{Cmpd}} \tau_{4(i-1)}^{ACT/360} \boxed{DF_{4i}}}{\sum_{i=1}^n \tau_{4(i-1)}^{ACT/360} \boxed{DF_{4i}}} \quad (5)$$

2. Libor-OIS Basis Swap($> 2Y$):

$$B^{LO} = \frac{S_{2n}^{3ML} \sum_{i=1}^{2n} \tau_{2(i-1)}^{30/360} \boxed{DF_{2i}} - \sum_{i=1}^{4n} \boxed{O_{i-1}^{Avg}} \tau_{i-1}^{ACT/360} \boxed{DF_i}}{\sum_{i=1}^{4n} \tau_{i-1}^{ACT/360} \boxed{DF_i}} \quad (6)$$

³Each par rate equation is calibrated to the market rate on the left-hand side by adjusting boxed variables. The sequence of each equation is same as curve building process in the M-Library.

3. 3M Libor Swap:

$$S^{3ML} = \frac{\sum_{i=1}^{4n} \boxed{L_{i-1}^{3M}} \tau_{i-1}^{ACT/360} DF_i}{\sum_{i=1}^{2n} \tau_{2(i-1)}^{30/360} DF_{2i}} \quad (7)$$

4. 3M-6M Basis Swap: (No Analytical Solution)

4 EUR Market

4.1 Product Equation

1. Overnight Index Swap($\leq 2Y$):

$$S^{OIS} \sum_{i=1}^n \tau_{4(i-1)}^{ACT/360} DF_{4i} = \sum_{i=1}^n O_{4(i-1)}^{Cmpd} \tau_{4(i-1)}^{ACT/360} DF_{4i} \quad (8)$$

2. Libor-OIS Basis Swap($> 2Y$):

$$\sum_{i=1}^n (O_{4(i-1)}^{Avg} + B^{LO}) \tau_{4(i-1)}^{ACT/360} DF_{4i} = \sum_{i=1}^{4n} L_{i-1}^{3M} \tau_{i-1}^{ACT/360} DF_i \quad (9)$$

3. 6M Libor Swap:

$$S^{6ML} \sum_{i=1}^n \tau_{4(i-1)}^{30/360} DF_{4i} = \sum_{i=1}^{2n} L_{2(i-1)}^{6M} \tau_{2(i-1)}^{ACT/360} DF_{2i} \quad (10)$$

4. 3M-6M Basis Swap⁴:

$$\sum_{i=1}^{4n} (L_{i-1}^{3M} + B^{36}) \tau_{i-1}^{ACT/360} DF_i = \sum_{i=1}^{2n} L_{2(i-1)}^{6M} \tau_{2(i-1)}^{ACT/360} DF_{2i} \quad (11)$$

5. Xccy Basis Swap(MtM):

$$\begin{aligned} & -DF_0^{Xccy} + \sum_{i=1}^{4n} (L_{i-1}^{3M} + B^{Xccy}) \tau_{i-1}^{ACT/360} DF_i^{Xccy} + DF_{4n}^{Xccy} \\ & = \sum_{i=1}^{4n} \frac{DF_{i-1}^{Xccy}}{DF_{i-1}^{USD}} (-DF_{i-1}^{USD} + L_{i-1}^{USD3M} \tau_{i-1}^{ACT/360} DF_i^{USD} + DF_i^{USD}) Cov_{i-1} \end{aligned} \quad (13)$$

6. Forward FX Constancy(Assumption):

$$FX_i = FX_0 \frac{DF_i^{Xccy}}{DF_i^{USD}} = FX_0 \frac{DF_i}{DF_i^{USD \text{ on EUR}}} \quad (14)$$

⁴The most liquid market is 3M-6M Basis from 2 Swaps, equations below.

$$\begin{cases} S^{3ML} \sum_{i=1}^n \tau_{4(i-1)}^{30/360} DF_{4i} = \sum_{i=1}^{4n} L_{i-1}^{3M} \tau_{i-1}^{ACT/360} DF_i \\ S^{6ML} \sum_{i=1}^n \tau_{4(i-1)}^{30/360} DF_{4i} = \sum_{i=1}^{2n} L_{2(i-1)}^{6M} \tau_{2(i-1)}^{ACT/360} DF_{2i} \\ B^{36} = S^{6ML} - S^{3ML} \end{cases} \quad (12)$$

4.2 Par Rate Equation

1. Overnight Index Swap($\leq 2Y$):

$$S^{OIS} = \frac{\sum_{i=1}^n \boxed{O_{4(i-1)}^{Cmpd}} \tau_{4(i-1)}^{ACT/360} \boxed{DF_{4i}}}{\sum_{i=1}^n \tau_{4(i-1)}^{ACT/360} \boxed{DF_{4i}}} \quad (15)$$

2. Libor-OIS Basis Swap($> 2Y$):

$$B^{LO} = \frac{S^{3ML} \sum_{i=1}^n \tau_{4(i-1)}^{30/360} \boxed{DF_{4i}} - \sum_{i=1}^n \boxed{O_{4(i-1)}^{Avg}} \tau_{4(i-1)}^{ACT/360} \boxed{DF_{4i}}}{\sum_{i=1}^n \tau_{4(i-1)}^{ACT/360} \boxed{DF_{4i}}} \quad (16)$$

3. 6M Libor Swap:

$$S^{6ML} = \frac{\sum_{i=1}^{2n} \boxed{L_{2(i-1)}^{6M}} \tau_{2(i-1)}^{ACT/360} DF_{2i}}{\sum_{i=1}^n \tau_{4(i-1)}^{30/360} DF_{4i}} \quad (17)$$

4. 3M-6M Basis Swap⁵:

$$B^{36} = \frac{\sum_{i=1}^{2n} L_{2(i-1)}^{6M} \tau_{2(i-1)}^{ACT/360} DF_{2i} - \sum_{i=1}^{4n} \boxed{L_{i-1}^{3M}} \tau_{i-1}^{ACT/360} DF_i}{\sum_{i=1}^{4n} \tau_{i-1}^{ACT/360} DF_i} \quad (18)$$

5. Xccy Basis Swap(MtM):

$$\begin{aligned} B^{Xccy} &= \frac{PV^{USD} - PV^{Xccy}}{\sum_{i=1}^{4n} \tau_{i-1}^{ACT/360} \boxed{DF_i^{Xccy}}}, \\ PV^{USD} &= \sum_{i=1}^{4n} \frac{\boxed{DF_{i-1}^{Xccy}}}{DF_{i-1}^{USD}} (-DF_{i-1}^{USD} + L_{i-1}^{USD3M} \tau_{i-1}^{ACT/360} DF_i^{USD} + DF_i^{USD}) Cov_{i-1}, \\ PV^{Xccy} &= -\boxed{DF_0^{Xccy}} + \sum_{i=1}^{4n} L_{i-1}^{3M} \tau_{i-1}^{ACT/360} \boxed{DF_i^{Xccy}} + \boxed{DF_{4n}^{Xccy}} \end{aligned} \quad (21)$$

⁵If we apply the 3M-6M Basis from 2 Swaps, the par rate equations are as follows.

$$S^{3ML} = \frac{\sum_{i=1}^{4n} \boxed{L_{i-1}^{3M}} \tau_{i-1}^{ACT/360} DF_{2i}}{\sum_{i=1}^n \tau_{4(i-1)}^{30/360} DF_{4i}} \quad (19)$$

$$B^{36} = \frac{\sum_{i=1}^{2n} L_{2(i-1)}^{6M} \tau_{2(i-1)}^{ACT/360} DF_{2i} - \sum_{i=1}^{4n} \boxed{L_{i-1}^{3M}} \tau_{i-1}^{ACT/360} DF_i}{\sum_{i=1}^n \tau_{4(i-1)}^{30/360} DF_{4i}} \quad (20)$$

6. Forward FX Constancy(Assumption):

$$\boxed{DF_i^{USD on EUR}} = \frac{DF_i^{USD}}{DF_i^{Xccy}} DF_i \quad (22)$$

$$i.e. \quad \boxed{DF_i^{[AAA] on EUR}} = \frac{DF_i^{[AAA] Xccy}}{DF_i^{Xccy}} DF_i \quad (23)$$

$$i.e. \quad \boxed{DF_i^{EUR on [AAA]}} = \frac{DF_i^{Xccy}}{DF_i^{USD}} DF_i^{USD on [AAA]} \quad (24)$$

5 JPY Market

5.1 Product Equation

1. Overnight Index Swap:

$$S^{OIS} \sum_{i=1}^n \tau_{4(i-1)}^{ACT/365} DF_{4i} = \sum_{i=1}^n O_{4(i-1)}^{Cmpd} \tau_{4(i-1)}^{ACT/365} DF_{4i} \quad (25)$$

2. 6M Libor Swap:

$$S^{6ML} \sum_{i=1}^{2n} \tau_{2(i-1)}^{ACT/365} DF_{2i} = \sum_{i=1}^{2n} L_{2(i-1)}^{6M} \tau_{2(i-1)}^{ACT/360} DF_{2i} \quad (26)$$

3. 3M-6M Basis Swap:

$$\sum_{i=1}^{4n} (L_{i-1}^{3M} + B^{36}) \tau_{i-1}^{ACT/360} DF_i = \sum_{i=1}^{2n} L_{2(i-1)}^{6M} \tau_{2(i-1)}^{ACT/360} DF_{2i} \quad (27)$$

4. Xccy Basis Swap(MtM):

$$\begin{aligned} & -DF_0^{Xccy} + \sum_{i=1}^{4n} (L_{i-1}^{3M} + B^{Xccy}) \tau_{i-1}^{ACT/360} DF_i^{Xccy} + DF_{4n}^{Xccy} \\ & = \sum_{i=1}^{4n} \frac{DF_{i-1}^{Xccy}}{DF_{i-1}^{USD}} (-DF_{i-1}^{USD} + L_{i-1}^{USD3M} \tau_{i-1}^{ACT/360} DF_i^{USD} + DF_i^{USD}) Cov_{i-1} \end{aligned} \quad (28)$$

5. Forward FX Constancy(Assumption):

$$FX_i = FX_0 \frac{DF_i^{USD}}{DF_i^{Xccy}} = FX_0 \frac{DF_i^{USD on JPY}}{DF_i} \quad (29)$$

5.2 Par Rate Equation

1. Overnight Index Swap:

$$S^{OIS} = \frac{\sum_{i=1}^n \boxed{O_{4(i-1)}^{Cmpd}} \tau_{4(i-1)}^{ACT/365} \boxed{DF_{4i}}}{\sum_{i=1}^n \tau_{4(i-1)}^{ACT/365} \boxed{DF_{4i}}} \quad (30)$$

2. 6M Libor Swap:

$$S^{6ML} = \frac{\sum_{i=1}^{2n} \boxed{L_{2(i-1)}^{6M}} \tau_{2(i-1)}^{ACT/360} DF_{2i}}{\sum_{i=1}^{2n} \tau_{2(i-1)}^{ACT/365} DF_{2i}} \quad (31)$$

3. 3M-6M Basis Swap:

$$B^{36} = \frac{\sum_{i=1}^{2n} L_{2(i-1)}^{6M} \tau_{2(i-1)}^{ACT/360} DF_{2i} - \sum_{i=1}^{4n} \boxed{L_{i-1}^{3M}} \tau_{i-1}^{ACT/360} DF_i}{\sum_{i=1}^{4n} \tau_{i-1}^{ACT/360} DF_i} \quad (32)$$

4. Xccy Basis Swap(MtM):

$$\begin{aligned} B^{Xccy} &= \frac{PV^{USD} - PV^{Xccy}}{\sum_{i=1}^{4n} \tau_{i-1}^{ACT/360} \boxed{DF_i^{Xccy}}}, \\ PV^{USD} &= \sum_{i=1}^{4n} \frac{\boxed{DF_{i-1}^{Xccy}}}{DF_{i-1}^{USD}} (-DF_{i-1}^{USD} + L_{i-1}^{USD3M} \tau_{i-1}^{ACT/360} DF_i^{USD} + DF_i^{USD}) Cov_{i-1}, \\ PV^{Xccy} &= -\boxed{DF_0^{Xccy}} + \sum_{i=1}^{4n} L_{i-1}^{3M} \tau_{i-1}^{ACT/360} \boxed{DF_i^{Xccy}} + \boxed{DF_{4n}^{Xccy}} \end{aligned} \quad (33)$$

5. Forward FX Constancy(Assumption):

$$\boxed{DF_i^{USD on JPY}} = \frac{DF_i^{USD}}{DF_i^{Xccy}} DF_i \quad (34)$$

$$i.e. \quad \boxed{DF_i^{[AAA] on JPY}} = \frac{DF_i^{[AAA] Xccy}}{DF_i^{Xccy}} DF_i \quad (35)$$

$$i.e. \quad \boxed{DF_i^{JPY on [AAA]}} = \frac{DF_i^{Xccy}}{DF_i^{USD}} DF_i^{USD on [AAA]} \quad (36)$$

6 AUD Market

6.1 Product Equation

1. Overnight Index Swap:

$$S^{OIS} \sum_{i=1}^n \tau_{4(i-1)}^{ACT/365} DF_{4i} = \sum_{i=1}^n O_{4(i-1)}^{Cmpd} \tau_{4(i-1)}^{ACT/365} DF_{4i} \quad (37)$$

2. 3M Libor Swap($\leq 3Y$):

$$S^{3ML} \sum_{i=1}^{4n} \tau_{i-1}^{ACT/365} DF_i = \sum_{i=1}^{4n} L_{i-1}^{3M} \tau_{i-1}^{ACT/365} DF_i \quad (38)$$

3. 6M Libor Swap($> 3Y$):

$$S^{6ML} \sum_{i=1}^{2n} \tau_{2(i-1)}^{ACT/365} DF_{2i} = \sum_{i=1}^{2n} L_{2(i-1)}^{6M} \tau_{2(i-1)}^{ACT/365} DF_{2i} \quad (39)$$

4. 3M-6M Basis Swap:

$$\sum_{i=1}^{4n} (L_{i-1}^{3M} + B^{36}) \tau_{i-1}^{ACT/365} DF_i = \sum_{i=1}^{2n} L_{2(i-1)}^{6M} \tau_{2(i-1)}^{ACT/365} DF_{2i} \quad (40)$$

5. Xccy Basis Swap(MtM):

$$\begin{aligned} & -DF_0^{Xccy} + \sum_{i=1}^{4n} (L_{i-1}^{3M} + B^{Xccy}) \tau_{i-1}^{ACT/360} DF_i^{Xccy} + DF_{4n}^{Xccy} \\ & = \sum_{i=1}^{4n} \frac{DF_{i-1}^{Xccy}}{DF_{i-1}^{USD}} (-DF_{i-1}^{USD} + L_{i-1}^{USD3M} \tau_{i-1}^{ACT/360} DF_i^{USD} + DF_i^{USD}) Cov_{i-1} \end{aligned} \quad (41)$$

6. Forward FX Constancy(Assumption):

$$FX_i = FX_0 \frac{DF_i^{Xccy}}{DF_i^{USD}} = FX_0 \frac{DF_i}{DF_i^{USD on AUD}} \quad (42)$$

6.2 Par Rate Equation

1. Overnight Index Swap:

$$S^{OIS} = \frac{\sum_{i=1}^n \boxed{O_{4(i-1)}^{Cmpd}} \tau_{4(i-1)}^{ACT/365} \boxed{DF_{4i}}}{\sum_{i=1}^n \tau_{4(i-1)}^{ACT/365} \boxed{DF_{4i}}} \quad (43)$$

2. 3M Libor Swap($\leq 3Y$):

$$S^{3ML} = \frac{\sum_{i=1}^{4n} \boxed{L_{i-1}^{3M}} \tau_{i-1}^{ACT/365} DF_i}{\sum_{i=1}^{4n} \tau_{i-1}^{ACT/365} DF_i} \quad (44)$$

3. 3M Libor Swap($> 3Y$)⁶:

$$S^{6ML} = \frac{\sum_{i=1}^{4n} (\boxed{L_{i-1}^{3M}} + B^{36}) \tau_{i-1}^{ACT/365} DF_i}{\sum_{i=1}^{2n} \tau_{2(i-1)}^{ACT/365} DF_{2i}} \quad (45)$$

4. 3M-6M Basis Swap:

$$B^{36} = \frac{\sum_{i=1}^{2n} \boxed{L_{2(i-1)}^{6M}} \tau_{2(i-1)}^{ACT/365} DF_{2i} - \sum_{i=1}^{4n} L_{i-1}^{3M} \tau_{i-1}^{ACT/365} DF_i}{\sum_{i=1}^{4n} \tau_{i-1}^{ACT/365} DF_i} \quad (46)$$

5. Xccy Basis Swap(MtM):

$$\begin{aligned} B^{Xccy} &= \frac{PV^{USD} - PV^{Xccy}}{\sum_{i=1}^{4n} \tau_{i-1}^{ACT/360} \boxed{DF_i^{Xccy}}}, \\ PV^{USD} &= \sum_{i=1}^{4n} \frac{\boxed{DF_{i-1}^{Xccy}}}{DF_{i-1}^{USD}} (-DF_{i-1}^{USD} + L_{i-1}^{USD3M} \tau_{i-1}^{ACT/360} DF_i^{USD} + DF_i^{USD}) Cov_{i-1}, \\ PV^{Xccy} &= -\boxed{DF_0^{Xccy}} + \sum_{i=1}^{4n} L_{i-1}^{3M} \tau_{i-1}^{ACT/360} \boxed{DF_i^{Xccy}} + \boxed{DF_{4n}^{Xccy}} \end{aligned} \quad (47)$$

6. Forward FX Constancy(Assumption):

$$\boxed{DF_i^{USD on AUD}} = \frac{DF_i^{USD}}{DF_i^{Xccy}} DF_i \quad (48)$$

$$i.e. \quad \boxed{DF_i^{[AAA] on AUD}} = \frac{DF_i^{[AAA] Xccy}}{DF_i^{Xccy}} DF_i \quad (49)$$

$$i.e. \quad \boxed{DF_i^{AUD on [AAA]}} = \frac{DF_i^{Xccy}}{DF_i^{USD}} DF_i^{USD on [AAA]} \quad (50)$$

⁶We substitute the 3M-6M Basis Swap equation into the 6M Libor Swap equation to calibrate L^{3M} in advance.

7 GBP (CHF) Market

Note⁷

7.1 Product Equation

1. Overnight Index Swap:

$$S^{OIS} \sum_{i=1}^n \tau_{4(i-1)}^{ACT/365} DF_{4i} = \sum_{i=1}^n O_{4(i-1)}^{Cmpd} \tau_{4(i-1)}^{ACT/365} DF_{4i} \quad (51)$$

2. 6M Libor Swap:

$$S^{6ML} \sum_{i=1}^{2n} \tau_{2(i-1)}^{ACT/365} DF_{2i} = \sum_{i=1}^{2n} L_{2(i-1)}^{6M} \tau_{2(i-1)}^{ACT/365} DF_{2i} \quad (52)$$

3. 3M-6M Basis Swap:

$$\sum_{i=1}^{4n} (L_{i-1}^{3M} + B^{36}) \tau_{i-1}^{ACT/365} DF_i = \sum_{i=1}^{2n} L_{2(i-1)}^{6M} \tau_{2(i-1)}^{ACT/365} DF_{2i} \quad (53)$$

4. Xccy Basis Swap(MtM):

$$\begin{aligned} & -DF_0^{Xccy} + \sum_{i=1}^{4n} (L_{i-1}^{3M} + B^{Xccy}) \tau_{i-1}^{ACT/360} DF_i^{Xccy} + DF_{4n}^{Xccy} \\ & = \sum_{i=1}^{4n} \frac{DF_{i-1}^{Xccy}}{DF_{i-1}^{USD}} (-DF_{i-1}^{USD} + L_{i-1}^{USD3M} \tau_{i-1}^{ACT/360} DF_i^{USD} + DF_i^{USD}) Cov_{i-1} \end{aligned} \quad (54)$$

5. Forward FX Constancy(Assumption):

$$FX_i = FX_0 \frac{DF_i^{Xccy}}{DF_i^{USD}} = FX_0 \frac{DF_i}{DF_i^{USD on GBP}} \quad (55)$$

7.2 Par Rate Equation

1. Overnight Index Swap:

$$S^{OIS} = \frac{\sum_{i=1}^n \boxed{O_{4(i-1)}^{Cmpd}} \tau_{4(i-1)}^{ACT/365} \boxed{DF_{4i}}}{\sum_{i=1}^n \tau_{4(i-1)}^{ACT/365} \boxed{DF_{4i}}} \quad (56)$$

⁷CHF also has the same curve building method as GBP but some market conventions (frequency, day count, etc.) are different.

2. 6M Libor Swap:

$$S^{6ML} = \frac{\sum_{i=1}^{2n} \boxed{L_{2(i-1)}^{6M}} \tau_{2(i-1)}^{ACT/365} DF_{2i}}{\sum_{i=1}^{2n} \tau_{2(i-1)}^{ACT/365} DF_{2i}} \quad (57)$$

3. 3M-6M Basis Swap:

$$B^{36} = \frac{\sum_{i=1}^{2n} L_{2(i-1)}^{6M} \tau_{2(i-1)}^{ACT/365} DF_{2i} - \sum_{i=1}^{4n} \boxed{L_{i-1}^{3M}} \tau_{i-1}^{ACT/365} DF_i}{\sum_{i=1}^{4n} \tau_{i-1}^{ACT/365} DF_i} \quad (58)$$

4. Xccy Basis Swap(MtM):

$$\begin{aligned} B^{Xccy} &= \frac{PV^{USD} - PV^{Xccy}}{\sum_{i=1}^{4n} \tau_{i-1}^{ACT/360} \boxed{DF_i^{Xccy}}}, \\ PV^{USD} &= \sum_{i=1}^{4n} \frac{\boxed{DF_{i-1}^{Xccy}}}{DF_{i-1}^{USD}} (-DF_{i-1}^{USD} + L_{i-1}^{USD3M} \tau_{i-1}^{ACT/360} DF_i^{USD} + DF_i^{USD}) Cov_{i-1}, \\ PV^{Xccy} &= -\boxed{DF_0^{Xccy}} + \sum_{i=1}^{4n} L_{i-1}^{3M} \tau_{i-1}^{ACT/360} \boxed{DF_i^{Xccy}} + \boxed{DF_{4n}^{Xccy}} \end{aligned} \quad (59)$$

5. Forward FX Constancy(Assumption):

$$\boxed{DF_i^{USD on GBP}} = \frac{DF_i^{USD}}{DF_i^{Xccy}} DF_i \quad (60)$$

$$i.e. \quad \boxed{DF_i^{[AAA] on GBP}} = \frac{DF_i^{[AAA] Xccy}}{DF_i^{Xccy}} DF_i \quad (61)$$

$$i.e. \quad \boxed{DF_i^{GBP on [AAA]}} = \frac{DF_i^{Xccy}}{DF_i^{USD}} DF_i^{USD on [AAA]} \quad (62)$$

8 ZAR (CAD, HKD, MXN, NZD, RUB, SGD) Market

Note^{8 9}

8.1 Product Equation

1. 3M Libor Swap:

$$S^{3ML} \sum_{i=1}^{4n} \tau_{i-1}^{ACT/365} DF_i^{3ML} = \sum_{i=1}^{4n} L_{i-1}^{3M} \tau_{i-1}^{ACT/365} DF_i^{3ML} \quad (63)$$

2. Xccy Basis Swap(MtM):

$$\begin{aligned} & -DF_0^{Xccy} + \sum_{i=1}^{4n} (L_{i-1}^{3M} + B^{Xccy}) \tau_{i-1}^{ACT/360} DF_i^{Xccy} + DF_{4n}^{Xccy} \\ & = \sum_{i=1}^{4n} \frac{DF_{i-1}^{Xccy}}{DF_{i-1}^{USD}} (-DF_{i-1}^{USD} + L_{i-1}^{USD3M} \tau_{i-1}^{ACT/360} DF_i^{USD} + DF_i^{USD}) Cov_{i-1} \end{aligned} \quad (64)$$

3. Forward FX Constancy(Assumption):

$$FX_i = FX_0 \frac{DF_i^{USD}}{DF_i^{Xccy}} = FX_0 \frac{DF_i^{USD on [AAA]}}{DF_i^{ZAR on [AAA]}} \quad (65)$$

8.2 Par Rate Equation

1. 3M Libor Swap:

$$S^{3ML} = \frac{\sum_{i=1}^{4n} \boxed{L_{i-1}^{3M}} \tau_{i-1}^{ACT/365} \boxed{DF_i^{3ML}}}{\sum_{i=1}^{4n} \tau_{i-1}^{ACT/365} \boxed{DF_i^{3ML}}} \quad (66)$$

⁸CAD, HKD, MXN, NZD, RUB and SGD also use the same curve building method as ZAR but some market conventions (frequency, day count, etc.) are different. None of them have OIS market.

⁹The following points are being considered.

1. The same curve building method as GBP will also apply to CAD, HKD, NZD, RUB and SGD if their OIS market has enough liquidity, and ZAR as well if we can get its economically reasonable OIS-like synthetic market.
2. Another curve building method written in [2] can apply to MXN if its Swap (Libor) market is USD collateral.

2. Xccy Basis Swap(MtM):

$$\begin{aligned}
B^{Xccy} &= \frac{PV^{USD} - PV^{Xccy}}{\sum_{i=1}^{4n} \tau_{i-1}^{ACT/360} \boxed{DF_i^{Xccy}}}, \\
PV^{USD} &= \sum_{i=1}^{4n} \frac{\boxed{DF_{i-1}^{Xccy}}}{DF_{i-1}^{USD}} (-DF_{i-1}^{USD} + L_{i-1}^{USD3M} \tau_{i-1}^{ACT/360} DF_i^{USD} + DF_i^{USD}) Cov_{i-1}, \\
PV^{Xccy} &= -\boxed{DF_0^{Xccy}} + \sum_{i=1}^{4n} L_{i-1}^{3M} \tau_{i-1}^{ACT/360} \boxed{DF_i^{Xccy}} + \boxed{DF_{4n}^{Xccy}}
\end{aligned} \tag{67}$$

3. Forward FX Constancy(Assumption):

$$\boxed{DF_i^{ZARon[AAA]}} = \frac{DF_i^{Xccy}}{DF_i^{USD}} DF_i^{USD on[AAA]} \tag{68}$$

9 TRY (KRW) Market

Note^{10 11}

9.1 Product Equation

1. 3M Libor Swap:

$$S^{3ML} \sum_{i=1}^n \tau_{4(i-1)}^{ACT/360} DF_{4i}^{3ML} = \sum_{i=1}^{4n} L_{i-1}^{3M} \tau_{i-1}^{ACT/360} DF_i^{3ML} \quad (69)$$

2. Forward FX($\leq 1Y$):

$$FX_i = FX_0 \frac{DF_i^{USD}}{DF_i^{Xccy}} \quad (70)$$

3. Xccy Swap($>1Y$, Non-MtM):

$$\begin{aligned} & -DF_0^{Xccy} + S^{Xccy} \sum_{i=1}^{4n} \tau_{i-1}^{ACT/360} DF_i^{Xccy} + DF_{4n}^{Xccy} \\ & = -DF_0^{USD} + \sum_{i=1}^{4n} L_{i-1}^{USD3M} \tau_{i-1}^{ACT/360} DF_i^{USD} + DF_{4n}^{USD} \end{aligned} \quad (71)$$

4. Forward FX Constancy(Assumption):

$$FX_i = FX_0 \frac{DF_i^{USD}}{DF_i^{Xccy}} = FX_0 \frac{DF_i^{USD \text{ on } [AAA]}}{DF_i^{TRY \text{ on } [AAA]}} \quad (72)$$

9.2 Par Rate Equation

1. 3M Libor Swap:

$$S^{3ML} = \frac{\sum_{i=1}^{4n} \boxed{L_{i-1}^{3M}} \tau_{i-1}^{ACT/360} \boxed{DF_i^{3ML}}}{\sum_{i=1}^n \tau_{4(i-1)}^{ACT/360} \boxed{DF_{4i}^{3ML}}} \quad (73)$$

2. Forward FX($\leq 1Y$):

$$\boxed{DF_i^{Xccy}} = \frac{FX_0}{FX_i} DF_i^{USD} \quad (74)$$

¹⁰KRW also has the same curve building method as TRY but some market conventions (frequency, day count, etc.) are different. None of them have OIS market, and they have Xccy Swap (fixed-floating) market which is more liquid than Xccy Basis Swap (floating-floating) market.

¹¹The GBP curve building method will also apply to KRW if its OIS market has enough liquidity, and TRY as well if we can get its economically reasonable OIS-like synthetic market.

3. Xccy Swap(>1Y, Non-MtM):

$$S^{Xccy} = \frac{-DF_0^{USD} + \sum_{i=1}^{4n} L_{i-1}^{USD3M} \tau_{i-1}^{ACT/360} DF_i^{USD} + DF_{4n}^{USD} + \boxed{DF_0^{Xccy}} - \boxed{DF_{4n}^{Xccy}}}{\sum_{i=1}^{4n} \tau_{i-1}^{ACT/360} \boxed{DF_i^{Xccy}}} \quad (75)$$

4. Forward FX Constancy(Assumption):

$$\boxed{DF_i^{TRYon[AAA]}} = \frac{DF_i^{Xccy}}{DF_i^{USD}} DF_i^{USD on[AAA]} \quad (76)$$

10 CNH (IDR, INR, MYR, PHP) Market

Note¹²

10.1 Product Equation

1. Forward FX($\leq 1Y$):

$$FX_i = FX_0 \frac{DF_i^{USD}}{DF_i^{Xccy}} \quad (77)$$

2. Xccy Swap($>1Y$, Non-MtM):

$$\begin{aligned} & -DF_0^{Xccy} + S^{Xccy} \sum_{i=1}^{4n} \tau_{i-1}^{ACT/360} DF_i^{Xccy} + DF_{4n}^{Xccy} \\ & = -DF_0^{USD} + \sum_{i=1}^{4n} L_{i-1}^{USD3M} \tau_{i-1}^{ACT/360} DF_i^{USD} + DF_{4n}^{USD} \end{aligned} \quad (78)$$

3. Forward FX Constancy(Assumption):

$$FX_i = FX_0 \frac{DF_i^{USD}}{DF_i^{Xccy}} = FX_0 \frac{DF_i^{USD_{Don}[AAA]}}{DF_i^{CNH_{on}[AAA]}} \quad (79)$$

10.2 Par Rate Equation

1. Forward FX($\leq 1Y$):

$$\boxed{DF_i^{Xccy}} = \frac{FX_0}{FX_i} DF_i^{USD} \quad (80)$$

2. Xccy Swap($>1Y$, Non-MtM):

$$S^{Xccy} = \frac{-DF_0^{USD} + \sum_{i=1}^{4n} L_{i-1}^{USD3M} \tau_{i-1}^{ACT/360} DF_i^{USD} + DF_{4n}^{USD} + \boxed{DF_0^{Xccy}} - \boxed{DF_{4n}^{Xccy}}}{\sum_{i=1}^{4n} \tau_{i-1}^{ACT/360} \boxed{DF_i^{Xccy}}} \quad (81)$$

3. Forward FX Constancy(Assumption):

$$\boxed{DF_i^{CNH_{on}[AAA]}} = \frac{DF_i^{Xccy}}{DF_i^{USD}} DF_i^{USD_{Don}[AAA]} \quad (82)$$

¹²IDR, INR, MYR and PHP also use the same curve building method as CNH but some market conventions (frequency, day count, etc.) are different. None of them have OIS or Libor market, and they have Xccy Swap market instead of Xccy Basis Swap market.

11 BRL (SEK, TWD) Market

Note¹³

11.1 Product Equation

1. Forward FX:

$$FX_i = FX_0 \frac{DF_i^{USD}}{DF_i^{Xccy}} \quad (83)$$

2. Forward FX Constancy(Assumption):

$$FX_i = FX_0 \frac{DF_i^{USD}}{DF_i^{Xccy}} = FX_0 \frac{DF_i^{USD on [AAA]}}{DF_i^{BRL on [AAA]}} \quad (84)$$

11.2 Par Rate Equation

1. Forward FX:

$$\boxed{DF_i^{Xccy}} = \frac{FX_0}{FX_i} DF_i^{USD} \quad (85)$$

2. Forward FX Constancy(Assumption):

$$\boxed{DF_i^{BRL on [AAA]}} = \frac{DF_i^{Xccy}}{DF_i^{USD}} DF_i^{USD on [AAA]} \quad (86)$$

¹³SEK and TWD also have the same curve building method as BRL but some market conventions (frequency, day count, etc.) are different. None of them have OIS, Libor or Xccy Swap market.

References

- [1] Masaaki Fujii, Yasufumi Shimada, Akihiko Takahashi, "A Note on Construction of Multiple Swap Curves with and without Collateral", p.15, Feb 2010.
- [2] Zhenyu Wu, "Use of CSA Curves for Pricing IRS Collateralized in a Foreign Currency", Bloomberg, Oct 2012.